

Lagrange's Mean-Value Theorem

Let $D \subseteq \mathbb{R}^n$ be a convex set and $a = (a_1, \dots, a_n)$,
 $b = (b_1, \dots, b_n)$ be distinct points in D .

T17. If $f: D \rightarrow \mathbb{R}$ possesses continuous partial derivatives on D , then $(\exists) \xi \in D$ such that

$$f(b) - f(a) = \nabla f(\xi) \cdot (b - a)$$

Proof: Define the auxiliary function $F: [0, 1] \rightarrow \mathbb{R}$, $F(t) = f(a + t(b - a))$.
By Lagrange's Mean-Value Theorem for functions of one variable, $(\exists) c \in (0, 1)$ s.t.

$$F(1) - F(0) = F'(c)$$

But $F(0) = f(a)$; $F(1) = f(b)$

$$\text{and } F'(t) = \underbrace{\frac{d}{dt} f(a_1 + t(b_1 - a_1), \dots, a_n + t(b_n - a_n))}_{\substack{\text{derivata} \\ \text{in raport cu } t}} \rightarrow \text{formula cunoscută} \\ a + t(b - a)$$

$$= \sum_{i=1}^n \frac{df}{dx_i}(a + t(b - a)) \underbrace{\frac{d}{dt}(a_i + t(b_i - a_i))}_{\substack{\uparrow \\ \text{inner product}}} \\ = \sum_{i=1}^n \frac{df}{dx_i}(a + t(b - a)) (b_i - a_i) = \nabla f(a + t(b - a)) \cdot (b - a)$$

Choosing $\xi = a + c(b - a)$, we obtain

$$f(b) - f(a) = \nabla f(\xi) \cdot (b - a) \quad \checkmark$$

Differential of a Function. Definitions

Let U and V be two normed linear spaces.

Def A function $f: U \rightarrow V$ is said to be Fréchet differentiable at a point $x_0 \in U$ if there exists a continuous linear mapping: $T: U \rightarrow V$ such that.

$$\lim_{x \rightarrow x_0} \frac{\|f(x) - f(x_0) - T(x - x_0)\|}{\|x - x_0\|} = 0.$$

The functional T is called the Fréchet differential of f at x_0 and is denoted by:
 $df(x_0)$

Differential of Functions of Several Variables

Let $D \subseteq \mathbb{R}^n$, $a \in \text{int}(D)$ and $f: D \rightarrow \mathbb{R}$ be a continuous function.

T19. If f has continuous partial derivatives at a , then it is differentiable at a

and

$$df(a)(h) = \nabla f(a) \cdot h, \quad \forall h \in \mathbb{R}^n \quad (*)$$